

Figure 5: Visualization of the marginals by varying iterations for EOFT. As iterations increase, the marginals calculated by the OFT-Sinkhorn algorithm (dotted line) gradually approach Ground Truth (black solid line).

Sparsity of the Coupling In domains such as logistics planning, an exact and sparse transport plan is desired. In this section, we conduct tests on the sparsity of the transport plan for EOFT-Sinkhorn methods with different regularization coefficients ϵ . We uniformly sample a network of 60 nodes, designating 15 as sources and 15 as targets. Figure 3 visualizes the transportation plan generated by the EOFT method. The dotted green lines represent potential node connections, whereas the red solid lines signify the actual transportation routes. The color intensity along these paths is indicative of the flow volume, offering a clear visual illustration of the plan’s density. The results reveal that as the of ϵ decrease, the approximate solution from the EOFT-Sinkhorn method converges more closely to the optimal solution achieved using Gurobi. Similarly, Figure 4 illustrates the transformation of the coupling matrix and marginals as the regularization coefficient ϵ decreased. With a decreasing ϵ , the optimal coupling matrix exhibits increased sparsity, and the transportation plan $\mathbf{P}$ progressively aligns with the solution delivered by Gurobi.

Numerical Stability Note the stability is critical in algorithm design and Figure 6 evaluates the numerical stability of the OFT-Sinkhorn algorithm. Each data point reflects the mean performance of four random instances, with all other parameters kept constant to ensure fair comparison. The heatmap displays both cost and time, where NaN represents a computational failure at that particular parameter combination. The horizontal axis shows the progressively reducing Err (the algorithm’s convergence threshold, defined as $\mathbf{P}\mathbf{1}_N - \mathbf{P}^\top\mathbf{1}_N - \mathbf{s}$) and the vertical axis illustrates the gradually decreasing ϵ . The findings reveal that as either the convergence threshold or ϵ diminishes, the EOFT algorithm requires more time to converge, yet it achieves a solution of higher quality. Our algorithm exhibits remarkable stability across a wide range of parameters, notably when ϵ range from 5×10^{-1} to 1×10^{-4} and convergence threshold ranges from 1×10^{-2} to 1×10^{-5} .

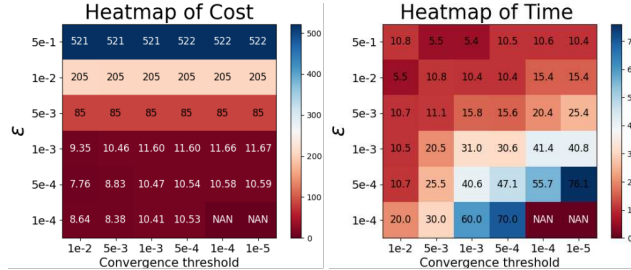


Figure 6: Visualization of the cost and time by varying the ϵ and Error(convergence threshold) for EOFT. Under most cases, EOFT exhibits a smooth changing process. Each point is obtained by the average of 4 instance

5 CONCLUSION AND FUTURE WORK

This paper has formulated the optimal flow transport problem beyond the vanilla discrete OT, to improve its applicability in real-world problems. We also present new time-efficient methods OFT-Sinkhorn algorithm to solve the problem and its variants, which is solved by GPU-friendly matrix iterations. Experiments show that our method is more efficient than general solvers in solving the minimum cost flow problem, which proves that our EOFT algorithm can be effectively used to solve the minimum cost flow problem. For future work, we will combine our approach with other OT variants, such as partial transportation Phatak et al. (2023), robust optimization Raghvendra et al. (2024), or explore parallel computation of OFT (Lahn et al., 2023).